

Spectral weight control in physics-informed neural networks: what the orthogonality penalty does, and how it transfers to shock-dominated shallow-water flow

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Abstract

Wang et al. [8] improve physics-informed neural network accuracy by factorising each hidden weight at initialisation, training the left factor and the singular values in place of the weight, and penalising departures of the left factor from orthogonality with a hand-tuned weight w_U . Their account assigns the gain to control over the weight spectrum. We reproduce their advection case over 20 seeds and then measure the trained factors, which their paper does not report. At the published $w_U = 1/35000$ the trained left factor is not close to orthogonal — $\|U^\top U - I\|_2$ has median 0.98, indicating a collapsed eigendirection — and the trained singular values differ from the effective weight’s actual singular values by up to a factor of 13. Constraining the factor to be exactly orthogonal, which removes w_U entirely, reaches only 1.7×10^{-2} median relative error against the penalty’s 8.4×10^{-3} ; removing the penalty from the same parameterisation returns the error to 2.1×10^{-1} , worse than an unconstrained network’s 8.4×10^{-2} . We read the term as a bound on the factor’s scale rather than an orthogonality constraint. Transferring the comparison to two-dimensional shallow-water dam breaks, where the solution carries a shock and the bounded-gradient hypothesis of the source paper’s Theorem 2.1 fails, no spectral regime separates from an unmodified network by more than about 10% in L^1 depth error on the three shock cases, and w_U can be moved three decades without measurable effect. On the one smooth initial condition in the set, exact orthogonality and spectrum-only training are a factor of 2.5 worse than an unmodified network. Establishing this required checking our finite-volume reference against an independently implemented solver family, which disagrees with it by more than the differences under test on two of three benchmarks — a floor that comparisons of this kind do not usually report.

1 Introduction

Physics-informed neural networks solve a PDE by minimising its residual at collocation points [5]. They fail on targets with steep or high-frequency structure, and a line of work attributes the failure to properties of the network’s Jacobian. Wang et al. [8] formalise this through a quantity they call the condition number, $\kappa(f) = \|J(f)\|_F$, and propose a remedy: decompose each hidden weight $W_j = U_j S_j V_j^\top$ once at initialisation, freeze V_j , train U_j and $\text{diag}(S_j)$ in place of W_j , and add

$$w_U \sum_j D_j^2, \quad D_j = \|U_j^\top U_j - I\|_2 \tag{1}$$

to the loss so that U_j stays orthogonal and $\text{diag}(S_j)$ therefore carries the weight’s spectrum.

Their conclusion lists three open problems. The penalty weight w_U is tuned per equation across two and a half decades, from 1/600 to 1/200000 over five PDEs, and their Section 4.6.3 reports that NTK- and DB-PINN-style adaptive weighting diverges on this term because the penalty starts at exactly zero. The penalty costs 1.28 to 1.83 times vanilla training time across their width and collocation sweeps. All five validation problems are, in their own description, controlled synthetic examples, and all five are smooth.

That last point has a specific consequence. Every hypothesis of their Theorem 2.1 requires $u \in H^1$ with $\|J(u)\|_F$ bounded. A dam-break solution carries a shock, so $\|J(u)\|_F$ is unbounded and $u \notin H^1$: the theorem is formally inapplicable to the entire problem class. The authors say as much in a different context, noting that the condition is violated once training collapses.

We take the three limitations in order and add the excluded regime. Section 4 reproduces the advection case and then measures what the penalty does to the trained factors, which decides what kind of constraint the term is. Section 5 moves the same four weight regimes to two-dimensional shallow-water dam breaks. Section 6.4 reports a result we did not set out to find: on two of three benchmarks the reference against which all of this is measured is less certain than the differences being measured.

2 The parameterisation and the four regimes compared

Write a hidden layer’s weight as $W \in \mathbb{R}^{m \times n}$ and its initial factorisation as $W_0 = U_0 S_0 V_0^\top$. All four regimes below keep $V = V_0$ frozen and differ in how U and the spectrum are treated.

Unconstrained. W trained directly. The baseline.

Penalised spectral. U and $s = \text{diag}(S)$ trained, with Eq. (1) added to the loss. This is the published method.

Exactly orthogonal. U constrained to the Stiefel manifold by the matrix-exponential parameterisation of Lezcano-Casado and Martínez-Rubio [3], and s trained. The penalty and w_U do not exist in this regime, which is what makes it the direct test of the first stated limitation.

Spectrum only. U and V both frozen at their orthogonal initial values, s alone trained. Structurally this layer is the one used by SVD-PINNs [2]; there it transfers a trained network to a new equation, whereas here it is trained from scratch. The construction is not ours and is not presented as new.

The distinction between the second and third regimes is the paper’s pivot. If the penalty is an approximation to orthogonality, then enforcing orthogonality exactly should do at least as well and costs no tuned weight. If it is something else, the two will separate.

3 Related work

Spectral structure in network weights. Spectral normalisation [4] divides a weight by its largest singular value to force a 1-Lipschitz map. It acts on the same object as the method studied here and with the opposite intent: it caps the gradient magnitude a network can express, whereas the diagnostic motivating [8] holds that networks fail on steep targets by expressing too little of it. The exactly-orthogonal regime we test uses the matrix-exponential parametrisation of Lezcano-Casado and Martínez-Rubio [3], introduced for recurrent networks; we are not aware of a prior application

to PINN training, and the result in Section 5 suggests why one might not have been reported. SVD-PINNs [2] freeze both orthogonal factors and retrain the singular values to move a trained network to a new equation. Our spectrum-only regime is that layer trained from scratch rather than transferred; the construction is theirs, and what we add is its failure mode as a from-scratch method.

Constraint handling in PINN losses. AL-PINNs [6] apply an augmented-Lagrangian treatment to the initial and boundary conditions — constraints on the solution, not on weight structure. The distinction matters here because the adaptive-weighting schemes that [8] report as diverging on their penalty are of this family, and the reason they diverge is specific to a penalty that starts at exactly zero.

PINNs at shocks. A separate line accepts that pointwise residuals are ill-defined at a discontinuity and changes the objective or the architecture: weak- and entropy-form losses, jump conditions imposed through the architecture, gradient-weighted losses concentrated near shocks, and relaxation formulations. None of these measures or controls a property of the network’s weights, so the comparison in Section 6 is orthogonal to them rather than competing with them; combining a spectral parametrisation with a weak-form loss is untested and outside what we report here. The shallow-water PINN literature specific to dam breaks is surveyed in [9], from which our benchmarks and finite-volume harness are taken.

What we did not find. A search in August 2026 returned no PINN work training weights under an exact orthogonality constraint, no condition-number or gradient-norm mismatch diagnostic for PINNs on hyperbolic problems, and no indexed citation of [8], which appeared online two months earlier. The searches were run against arXiv-indexed material and are not exhaustive; the sweep needs repeating before submission.

4 Implementation

The penalty in Eq. (1) uses the matrix 2-norm, following the source paper. An early version of our code used the Frobenius norm, which at width 128 is up to 128 times stronger. Both are reported: the dam-break matrix of Table 3 was produced with the Frobenius penalty and the sweep of Table 4 with Eq. (1), and Section 6.3 compares them directly. The choice turns out to leave integrated depth error alone and to change where the shock is placed. All results here use L-BFGS at 3000 iterations for the advection case and Adam at 10^{-3} for 20000 iterations on the shallow-water cases. Seeds are set before model construction, so initial weights are reproducible.

5 Reproduction and mechanism on the advection case

5.1 The gate

Table 1 repeats the source paper’s Section 4.1 advection setup: a 3×25 tanh network, Xavier initialisation, 100 boundary and 500 interior points by Latin hypercube sampling, L-BFGS to 3000 iterations, $w_U = 1/35000$. We report 20 seeds. Distributions in this problem are heavy-tailed — the unconstrained network’s worst seed is 53 times its best — so the median is the more stable summary and both are given.

Table 1: Relative L^2 error on the advection case over 20 seeds. The first two rows are the comparison the source paper reports (their Table 5, tanh activation); the last three are regimes it does not test. Reproducing the published cells is what licenses reading the rest.

Regime	Mean $\pm$ s.d.	Median	Wang et al. Table 5
Unconstrained	$(1.37 \pm 1.4) \times 10^{-1}$	8.4×10^{-2}	$(1.13 \pm 1.67) \times 10^{-1}$
Penalised, $w_U = 1/35000$	$(1.46 \pm 1.5) \times 10^{-2}$	8.4×10^{-3}	$(5.64 \pm 3.86) \times 10^{-3}$
Exactly orthogonal	$(3.31 \pm 3.9) \times 10^{-2}$	1.7×10^{-2}	—
Spectrum only	$(3.29 \pm 5.5) \times 10^{-1}$	1.2×10^{-1}	—
Same parameterisation, $w_U = 0$	$(3.37 \pm 3.5) \times 10^{-1}$	2.1×10^{-1}	—

The two published cells reproduce within their reported spread, so the setup is faithful. Three readings follow from the rest. Exact orthogonality, which eliminates w_U , costs about a factor of two in median error against the tuned penalty. Training the spectrum alone is worse than an unconstrained network and diverges on 2 of 20 seeds, so the trained directions rather than the trained spectrum carry the gain — a negative result for the cheap variant, and one that already sits awkwardly with a spectral explanation of the mechanism. Removing the penalty while keeping the parameterisation is the worst regime of the five.

5.2 What the penalty does to the trained factors

The source paper reports errors, not factor geometry. We measure, per factorised layer of trained networks over 3 seeds, the orthogonality defect $D = \|U^\top U - I\|_2$ and the largest relative gap between the trained $|s|$ and the singular values of the effective weight $USV^\top$ (Table 2).

Table 2: Factor geometry after training, 3 seeds, median over layers. If the described mechanism held, D would be near zero and the two spectra would coincide.

Setting	Median error	Median D	$ s $ vs. true singular values
$w_U = 1/35000$	7.9×10^{-3}	0.98	up to 13 $\times$ apart
$w_U = 0$	3.05×10^{-1}	63 (max 177)	up to 31 $\times$ apart

At the published weight $D \approx 1$, which for a defect measured in the 2-norm means an eigendirection of $U^\top U$ has collapsed: the trained factor is rank-deficient, not near-orthogonal. The trained s is not the effective weight’s spectrum. Both premises of the stated mechanism fail, while the method’s accuracy advantage holds.

Ordering the four regimes by median error gives unconstrained-parameterised 2.1×10^{-1} , unconstrained 8.4×10^{-2} , exactly orthogonal 1.7×10^{-2} , weakly penalised 8.4×10^{-3} . The penalty is doing something the method needs; what it is doing is bounding the scale of U , since removing it lets the defect grow by two orders of magnitude and destroys the gain, while enforcing the constraint it nominally approximates recovers only about half of it. Exact orthogonality is a stronger constraint than the problem wants.

This reframes the first stated limitation. The open question is not how to adapt w_U but which constraint class the term belongs to, and the ordering above predicts that a bound weaker than orthogonality — a spectral-norm cap on U , or an explicit box on its singular values — should recover the gain without a tuned penalty. Testing that prediction is the next step, and it is not yet done.

The evidence here is one smooth one-dimensional problem, one architecture and one optimiser. Whether the ordering survives on a shock is a separate question, which is what Section 5 asks.

6 Shallow-water dam breaks

6.1 Setup

The benchmarks are two-dimensional dam breaks over a Gaussian-hump bed at the conventions of the comparative study of Zahra et al. [9]: domain $[0, 100]^2$ m, $g = 2$ m/s², reflective walls on all four sides, $t_{\text{end}} = 2$ s. We use a step release, a circular column over a wet bed, the same column over a dry bed, and a smooth Gaussian mound, all with Manning $n = 0$. A fifth case, three submerged humps with Manning friction over 20 s, is included and then set aside for the reason given below. Errors are L^1 and L^2 of depth at the final output time on a shared 128^2 evaluation grid, against a reference computed in-process with an HLLC MUSCL–van Leer finite-volume scheme [7] at 512^2 with hydrostatic reconstruction for well-balancing [1]. Baselines are an unmodified network, a Fourier-feature network — the standard remedy for high-frequency targets, absent from the source paper’s comparison — and a finite-volume-informed network. Five seeds per cell.

6.2 Results

Table 3 gives all five benchmarks, five seeds each. On the three shock-dominated cases no spectral regime separates from an unmodified network by more than about 10% in L^1 depth: on the step release every entry but the finite-volume-informed one falls between 1.03 and 1.13×10^3 , inside the unconstrained baseline’s own seed spread. The unconstrained network sits a factor of 4.1 to 4.4 above a classical scheme at the same resolution.

Table 3: L^1 depth error at the final output time on the 128^2 evaluation grid, mean $\pm$ s.d. over 5 seeds. The classical row is HLLC MUSCL–van Leer at the evaluation resolution, not the reference. Section 6.4 gives the reference uncertainty for the first, third and fourth columns; it exceeds several of the differences in this table.

Entry	Circular, wet	Circular, dry	Step	Gaussian mound	Three humps
Classical, 128^2	3.914×10^2	3.798×10^2	2.476×10^2	6.357×10^0	7.965×10^0
Unconstrained	$(1.602 \pm 0.012) \times 10^3$	$(1.681 \pm 0.076) \times 10^3$	$(1.028 \pm 0.11) \times 10^3$	$(3.83 \pm 0.22) \times 10^2$	$(8.87 \pm 0.6) \times 10^2$
Fourier features	$(1.681 \pm 0.11) \times 10^3$	$(1.838 \pm 0.12) \times 10^3$	$(1.124 \pm 0.068) \times 10^3$	$(4.03 \pm 0.48) \times 10^2$	$(7.96 \pm 1.5) \times 10^2$
Penalised spectral	$(1.662 \pm 0.12) \times 10^3$	$(1.602 \pm 0.054) \times 10^3$	$(1.106 \pm 0.047) \times 10^3$	$(4.69 \pm 0.55) \times 10^2$	$(7.49 \pm 1.1) \times 10^2$
Exactly orthogonal	$(1.725 \pm 0.13) \times 10^3$	$(1.507 \pm 0.14) \times 10^3$	$(1.054 \pm 0.071) \times 10^3$	$(9.69 \pm 1.8) \times 10^2$	$(9.19 \pm 2.3) \times 10^2$
Spectrum only	$(2.347 \pm 0.11) \times 10^3$	$(2.314 \pm 0.19) \times 10^3$	$(1.126 \pm 0.015) \times 10^3$	$(9.16 \pm 0.34) \times 10^2$	$(6.08 \pm 1.1) \times 10^2$
FV-informed	$(1.481 \pm 0.24) \times 10^4$	$(0.932 \pm 0.054) \times 10^4$	$(1.365 \pm 0.27) \times 10^4$	$(4.40 \pm 0.56) \times 10^3$	$(8.66 \pm 0.2) \times 10^2$

The two benchmarks that do separate separate against the method. On the Gaussian mound — the only smooth initial condition in the set, and the case closest to the regime the source paper validates in — exact orthogonality reaches $(9.69 \pm 1.8) \times 10^2$ against the unconstrained network’s $(3.83 \pm 0.22) \times 10^2$, and spectrum-only training $(9.16 \pm 0.34) \times 10^2$. That is a factor of 2.5 in the wrong direction, it is far outside the seed spread, and Section 6.4 puts the reference uncertainty on this benchmark at 3.8×10^2 , so the gap is resolvable where smaller ones are not. The penalised regime is intermediate at $(4.69 \pm 0.55) \times 10^2$. Constraining the factor hurts most where the target is smooth.

On the three-hump case with Manning friction the ordering inverts: spectrum-only training is the best neural entry at $(6.08 \pm 1.1) \times 10^2$ and exact orthogonality the worst at $(9.19 \pm 2.3) \times 10^2$. We do not read this as support for the spectral family, because every entry on that benchmark loses between 63% and 90% of its initial mass over the 20 s run — against 10^{-3} or below on the two-second cases — so all six models are far outside any regime where a depth error is meaningful. The case is reported for completeness and should not be used to rank methods.

The finite-volume-informed network collapses on the four short cases, matching the behaviour reported for it in [9].

6.3 Does the penalty weight matter here?

The first of the source paper’s stated limitations is that w_U must be tuned per equation, and no published value exists for shallow-water flow. We bracketed it over three decades on the two circular benchmarks with the corrected Eq. (1) penalty, five seeds per cell (Table 4).

Table 4: L^1 depth error against w_U over three decades, five seeds, using the 2-norm penalty of Eq. (1). The spread across the whole range is smaller than the spread across seeds within any one cell.

w_U	Circular, wet bed	Circular, dry bed
10^{-3}	$(1.578 \pm 0.025) \times 10^3$	$(1.765 \pm 0.032) \times 10^3$
$1/35000 = 2.9 \times 10^{-5}$	$(1.601 \pm 0.12) \times 10^3$	$(1.709 \pm 0.056) \times 10^3$
10^{-6}	$(1.549 \pm 0.069) \times 10^3$	$(1.643 \pm 0.050) \times 10^3$

Three orders of magnitude in w_U move the error by 3% on the wet bed and 7% on the dry bed, both inside the seed-to-seed spread. On these problems the tuning burden the source paper identifies does not exist, because the knob does not do anything. That is a negative answer to their first limitation in this regime rather than a solution to it: there is nothing to adapt. It also says the penalty is not the active ingredient here, which is consistent with the whole spectral family sitting within 10% of an unmodified network on the same benchmarks.

This sweep supersedes one reading we drew from the earlier, Frobenius-penalty run and now retract. There, the penalised network placed the radial front on the dry bed at 3.5 ± 4.1 m from the reference against the unconstrained network’s 21.2 ± 8.6 m, which we took as a local advantage invisible to the integrated error. Under the corrected penalty the same quantity is 14.8 ± 6.6 m at $w_U = 1/35000$ and between 14.8 and 17.0 m across the three decades, overlapping the unconstrained baseline within one standard deviation. The apparent front-position advantage belonged to the stronger Frobenius penalty, not to the published method, and it does not survive. Integrated depth error was almost unaffected by the same change (1.662×10^3 against 1.601×10^3 on the wet bed), so the norm choice matters for where the shock is placed and not for the depth field as a whole.

6.4 How far the reference can be trusted

Every number in Table 3 is measured against one solver’s output. To find out what that is worth, we compare the reference against an independent implementation: the HLL, Lax–Wendroff and MUSCL–Rusanov runs produced for [9] on a 501^2 node grid for the same initial conditions, interpolated onto the evaluation grid. Those runs store depth only, so the comparison is on depth.

Three regimes appear. On the wet-bed column the reference is vindicated: it agrees with an independent HLL implementation to 3.11×10^2 , tighter than its own 128^2 discretisation error of

Table 5: L^1 depth disagreement between our 512² HLLC reference and three independently implemented schemes on the same initial condition, against our own discretisation error and the unconstrained network’s error. Read down a column: where the middle entries exceed the last, the benchmark cannot resolve differences of the size Table 3 reports.

	Circular, wet bed	Gaussian mound	Step release
Our classical at 128 ²	3.914×10^2	6.357×10^0	2.476×10^2
vs. their HLL	3.111×10^2	3.763×10^2	2.169×10^3
vs. their MUSCL–Rusanov	7.395×10^2	3.949×10^2	2.208×10^3
vs. their Lax–Wendroff	7.551×10^3	3.435×10^3	9.349×10^3
Unconstrained network	1.602×10^3	$\sim 3.9 \times 10^2$	1.028×10^3

3.91×10^2 , and both are far below the 1.60×10^3 network error. Differences between network variants on that benchmark are real. On the Gaussian mound the two solver families disagree by 3.76×10^2 and the network error is 3.9×10^2 — the same number — while our own discretisation error is sixty times smaller at 6.36. On the step release the solver families differ by 2.2×10^3 , twice the network error. No ranking of network variants on those two benchmarks survives a change of reference solver.

Lax–Wendroff is the outlier throughout, which is expected of Lax–Wendroff with artificial viscosity at a shock and is a statement about that scheme rather than about ours. Nothing here says our reference is wrong; it is the most refined and least diffusive of the four, and on the wet-bed column it is confirmed. What it says is that a comparison table of this kind has a resolution floor that varies by benchmark, and that the floor has to be quoted next to any margin claimed from it. We have not seen this check performed in the PINN-for-conservation-laws literature, where a single fine-grid run is the usual reference.

7 Limitations

Five seeds per cell resolve a factor of two, not the 10% differences that make up most of Table 3; only the Gaussian-mound column separates by more than the spread. Section 6.4 bounds what the comparison can resolve at all and should be read before Table 3, not after it. The three-hump case is reported but not interpretable: every model there loses most of its mass.

The factor-geometry measurement of Table 2 uses 3 seeds on one one-dimensional problem, one architecture and one optimiser, and it is the basis for the paper’s central claim about what the penalty does. It should be repeated on the shallow-water networks, where the per-run factor diagnostics are now recorded but not yet analysed.

The constraint that Section 5 predicts — a bound on the factor’s largest singular value, weaker than orthogonality and carrying no tuned weight — is implemented and running at the time of writing. Until it reports, the prediction stands as a prediction.

Two of the five contributions in the project proposal have no section here. The pointwise condition-number field, which is the diagnostic that would explain *where* on the shock the spectral variants fail rather than only that they do, is not built. Neither is the Navier–Stokes leg — the Taylor–Green case the source paper reports and the lid-driven cavity — which is what would connect these results back to the setting [8] validates in. Cost and memory were recorded per run but are not tabulated, so the second stated limitation of the source paper is not yet answered quantitatively.

8 Conclusion

The orthogonality penalty of [8] is load-bearing on their own advection problem and is not doing what the paper says it does. At the published weight the trained factor is rank-deficient rather than orthogonal, and the trained singular values are not the effective weight’s spectrum, yet removing the penalty destroys the accuracy gain and enforcing exact orthogonality recovers only about half of it. The term acts as a bound on the factor’s scale.

Moving to shock-dominated shallow-water flow, outside the reach of the source paper’s Theorem 2.1, the picture is flatter and in one place reversed. On the three shock cases no spectral regime separates from an unmodified network by more than about 10% in integrated depth error, and the penalty weight can be moved three decades without measurable effect, so the tuning problem the source paper poses does not arise. On the one smooth case in the set, exact orthogonality and spectrum-only training are a factor of 2.5 *worse* than an unmodified network, by a margin that exceeds the reference uncertainty. Whatever the reparameterisation contributes on smooth one-dimensional targets does not transfer, and constraining the factor hurts most where the target is smoothest.

Establishing any of this required checking the reference against an independent solver family, which on two of three benchmarks disagrees with it by more than the differences under test. We would encourage that check as routine practice: a single fine-grid run is the usual reference in this literature, and its uncertainty is rarely quoted.

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